

# Fine-Grained Fruit Classification

*EfficientNet-B3 Classification*

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# 1. Our topic: Fine-grained Fruit Classification

Our goal is to build a system that classifies 20 fruit varieties from photographs.

|                                                                                                                |                                                                                                                 |                                                                                                                 |                                                                                                                 |                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|  Apples<br><b>8 varieties</b> |  Bananas<br><b>3 varieties</b> |  Kiwis<br><b>2 varieties</b> |  Pears<br><b>5 varieties</b> |  Tomatoes<br><b>2 varieties</b> |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|



The core challenge: many fruits look nearly identical.

## 2. Dataset

### How Images Were Captured

- Devices:** iPhone & Android smartphones
- Backgrounds:** Tables, bags, hands, various surfaces
- Angles:** Top, side, close-up, and angled shots
- Lighting:** Natural, artificial, and mixed light
- Ripeness:** Multiple ripeness stages per variety
- Packaging:** Some fruits in clear plastic bags

*32 robot camera images for each apple royal gala, pink lady, granny smith kept separate for fine tuning and evaluation.*



### 3. Why EfficientNet-B3?

Most CNN architectures scale up by going either

#### DEEPER (more layers)

- ✓ Early layers learn edges → middle layers learn shapes → deep layers learn high-level concepts
- ✗ Vanishing gradients at extreme depth; sharp diminishing returns

#### WIDER (more channels)

- ✓ More diverse feature detectors running in parallel per layer
- ✗ Saturates fast. A very wide but shallow network can't learn complex multi-step abstractions, because width doesn't add representational depth.

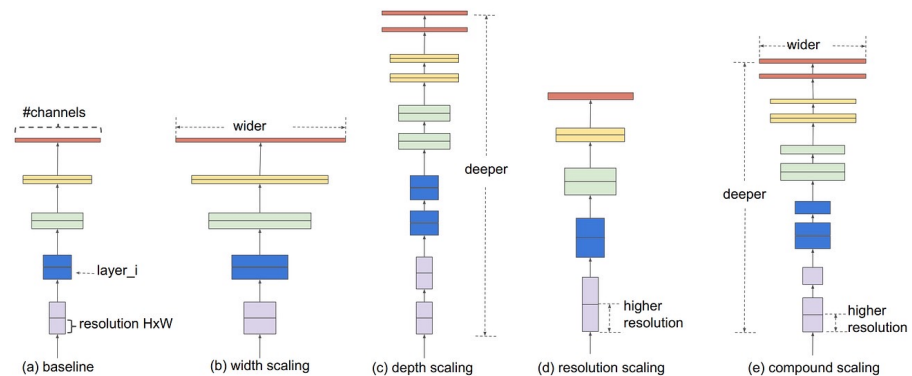
#### HIGHER RESOLUTION (larger input)

- ✓ More spatial detail — catches subtle texture & colour differences
- ✗ 4× more compute per resolution doubling; accuracy plateaus fast

#### EfficientNetB3 follows compound scaling

It scales all three together in a principled ratio, keeping the network balanced.

- Higher resolution is useless if the network is too shallow to process the detail.
- Going deeper is wasteful if channels are too narrow to carry information.



Tan & Le (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ICML.

### 3. Why EfficientNet-B3?

#### Why EfficientNet-B3?

##### 300 × 300 input

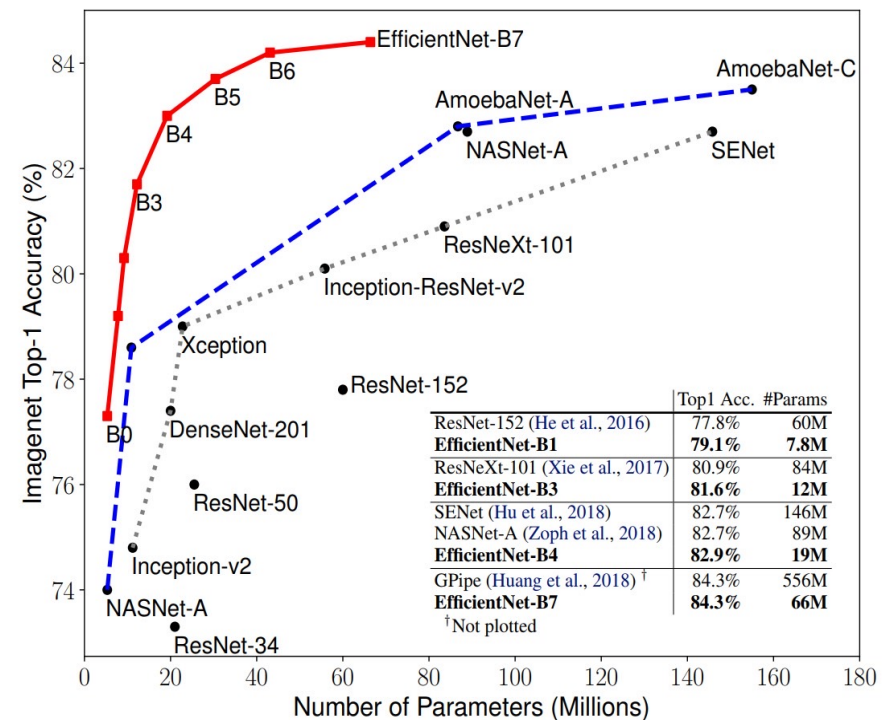
High-resolution input captures the fine colour & texture detail needed to tell apple varieties apart.

##### 12M parameters

Compact enough to train well on a small dataset of ~1,500 images.

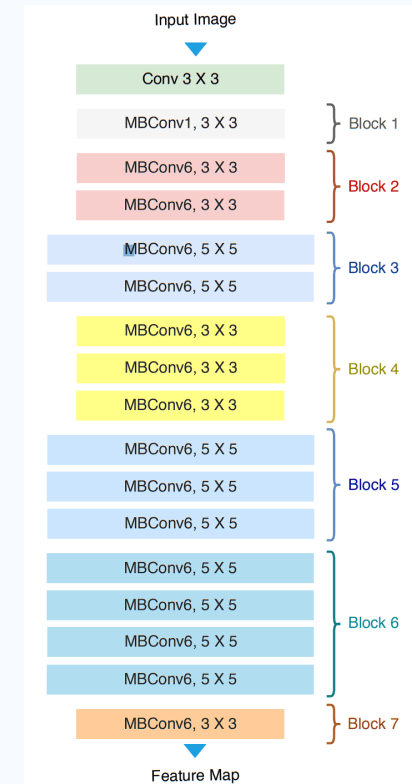
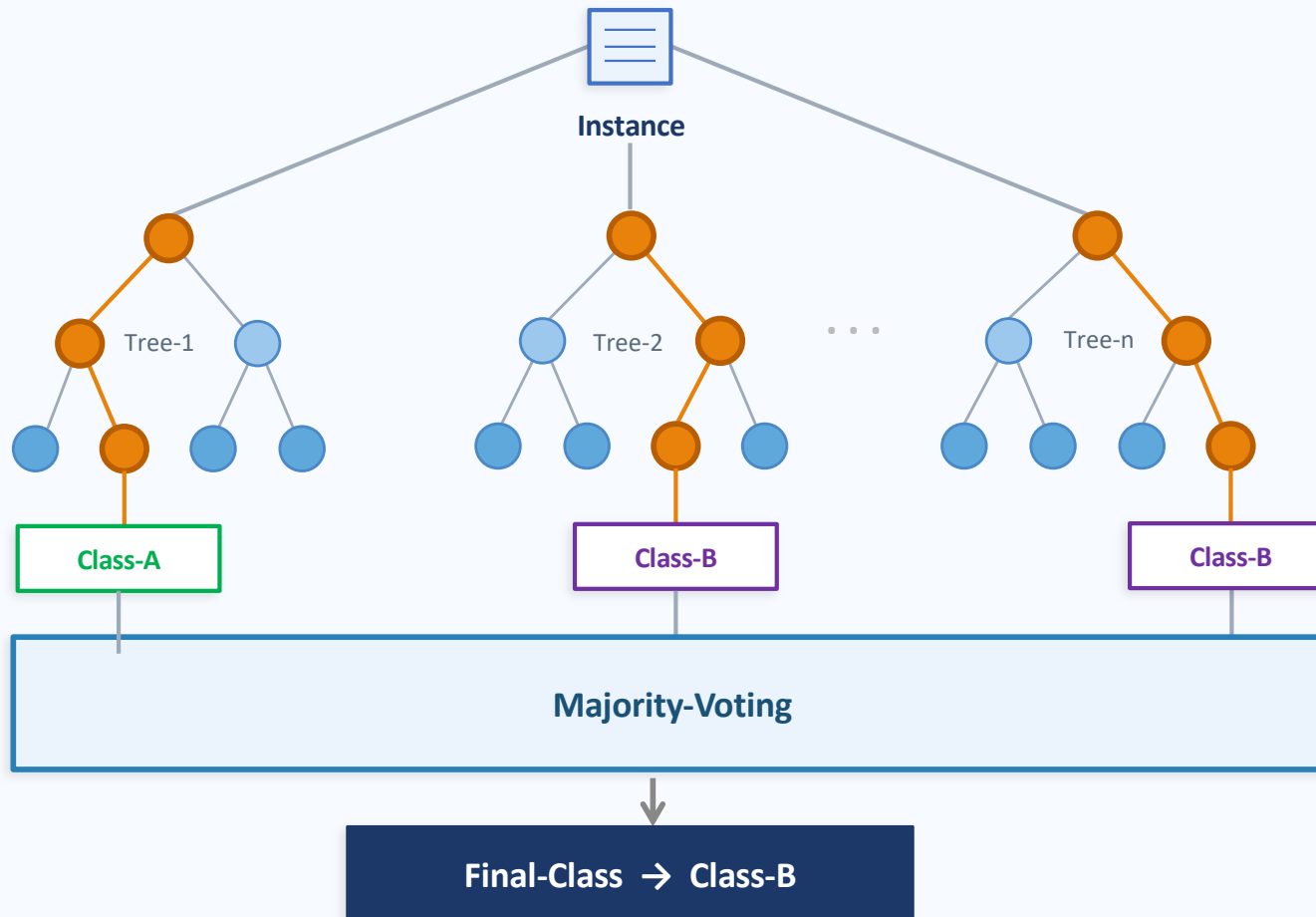
##### ImageNet pretrained

Starts with rich general visual knowledge, needing only fine-tuning for fruit.



Tan & Le (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ICML.

## 4. EfficientNet-B3 Architecture: Inspiration



Tashin & Noor (2021). EfficientNet: Classification and understanding of cloud structures via satellite images with EfficientUNet

## 4. Tapping Mechanism



## 5. Training Technique

### Stage 1 — Foundation Training

- Train on original phone images (30 epochs)
- Frozen backbone first, then top 3 blocks unfrozen
- Goal: build reliable fruit features
- Generate Grad-CAM heat maps



### Stage 2 — Guided Augmentation

- Use heat maps to apply smart augmentation:
  - Blur backgrounds outside the fruit region
  - Crop & enlarge the high-attention fruit area
- Train for another 30 epochs
- Top 4 backbone blocks unfrozen



### Phase 3 — Robot Fine-Tuning

- Start from best Stage 2 checkpoint
- 15 epochs on ~20 robot images per class
- Adapts to new camera environment quickly
- Robot images kept fully separate until here

**Why two stages?** The model must first learn to classify fruit before its attention maps become reliable enough to guide augmentation. Applying Grad-CAM guidance from scratch would be circular — the model has nothing meaningful to attend to yet.



## 6. Results: Standard 20-Class Performance

**91.55%**

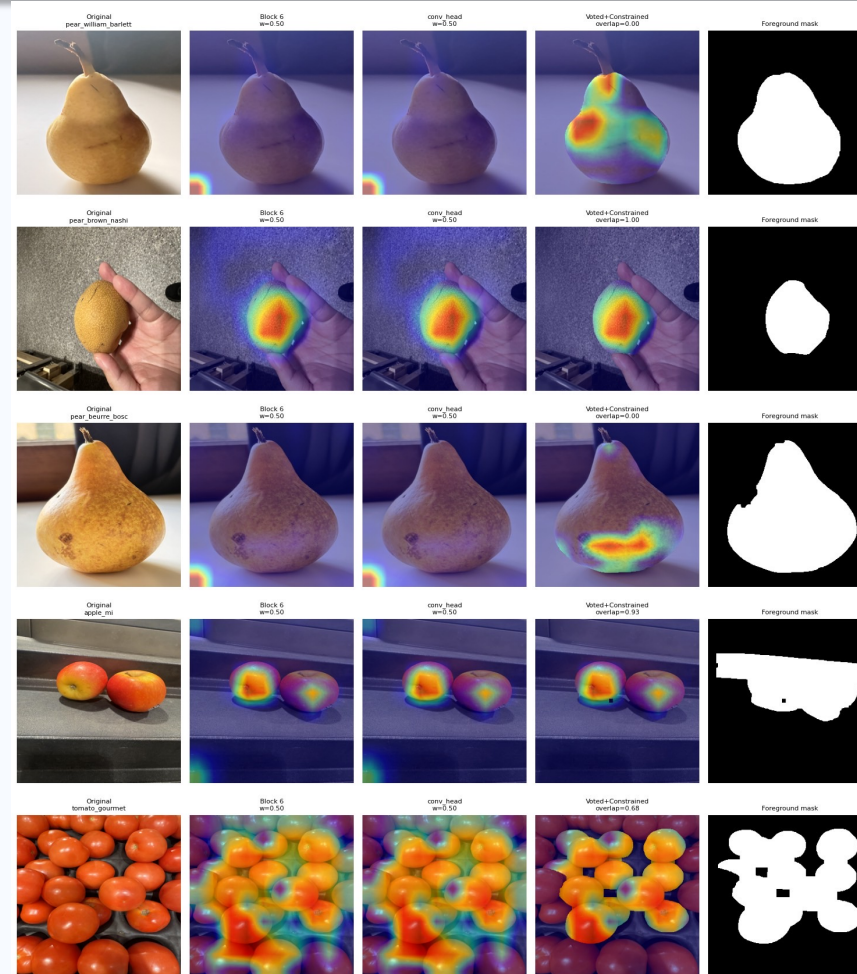
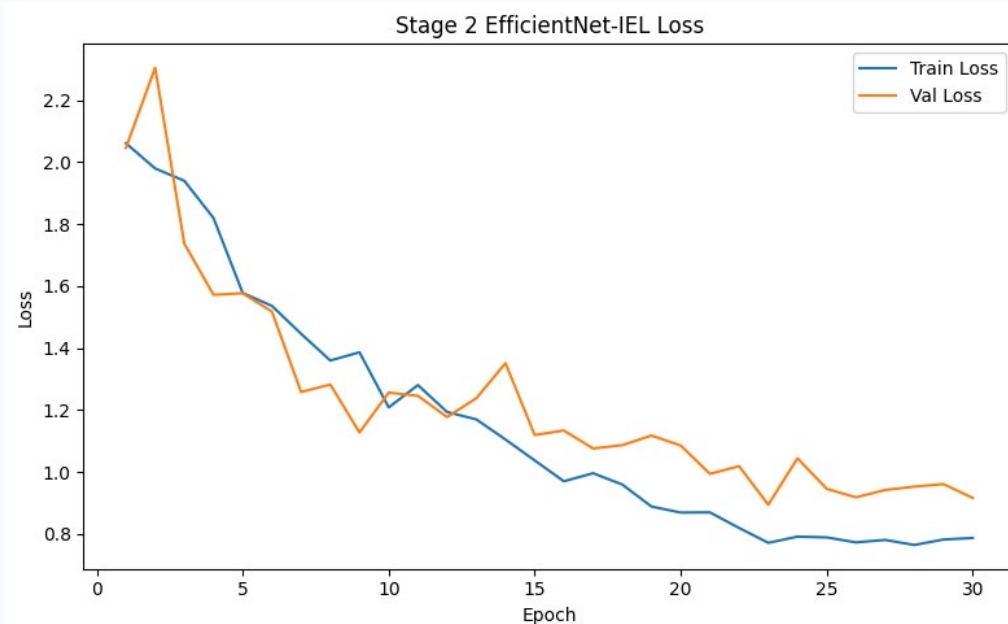
Test Accuracy

426 images

**0.9098**

Macro F1 Score

Across all 20 classes



## 6. Results: Robot Camera Evaluation

### Before Fine-Tuning

```
=====
PHASE 3 - PRE-FINETUNE: ROBOT TEST SET EVALUATION
=====
              precision    recall  f1-score   support

apple_grannysmith      1.0000      1.0000      1.0000         8
apple_pinklady          1.0000      0.3750      0.5455         8
apple_royalgala         0.0000      0.0000      0.0000         8

   micro avg           1.0000      0.4583      0.6286        24
   macro avg           0.6667      0.4583      0.5152        24
  weighted avg           0.6667      0.4583      0.5152        24
```



### After Fine-Tuning

```
=====
PHASE 3 - POST-FINETUNE: ROBOT TEST SET EVALUATION
=====
              precision    recall  f1-score   support

apple_grannysmith      1.0000      1.0000      1.0000         8
apple_pinklady          1.0000      1.0000      1.0000         8
apple_royalgala         1.0000      1.0000      1.0000         8

   accuracy              1.0000              1.0000        24
   macro avg             1.0000      1.0000      1.0000        24
  weighted avg             1.0000      1.0000      1.0000        24
```

*60/15/25 Train-Test Split - Fine Tune for 15 epochs*

## 7. Discussion: What Worked & Common Errors

### ✓ What Worked Well

#### Grad-CAM guided augmentation

Stage 2 consistently lowered validation loss across all 30 epochs — genuine generalisation gain.

#### Multi-layer voting

Generally, the architecture works well, it captures on the fruit.

#### EfficientNet compound scaling

Rich, multi-level feature hierarchy captured both colour and texture detail.

#### Rapid robot adaptation

High accuracy on robot test set after just 15 fine-tuning epochs with minimal data.

### ✗ Common Error Patterns

#### Multi-layer voting

Although it works well, it still focuses on backgrounds in some images. This technique should be more clearly examine to provide higher accuracy

#### Unusual capture conditions and more images

Lighting is one of main issues as it changes the fruits colors which can affect the prediction. More images with different lighting setting could help the improvement. Similar issue to fruits in plastic bags. Require a technique to separate the bag and the object inside of it.

#### Robot images

Images from robot has lower resolution, and the environment setting is quite dark which makes it hard to differentiate apple pink lady and apple royal gala.

**Thank You**